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Deep Learning Techniques for Text Summarization Using Recurrent Neural Networks

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by

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ABSTRACT

Condensing a lengthy text into a shorter form while maintaining its essential meaning is the goal of text summarization, a crucial work in natural language processing (NLP). With an emphasis on Bidirectional Long Short-Term Memory (BiLSTM) networks and attention processes, this dissertation investigates the use of deep learning approaches for text summarization utilizing recurrent neural networks (RNNs).

To handle tokenized and padded input text sequences, the suggested model uses a sequence-to-sequence (Seq2Seq) architecture with a BiLSTM encoder and a decoder. In order to create a combined context vector that captures the syntactic and semantic subtleties of the input sequence, the encoder makes use of the BiLSTM's capacity to recognize both forward and backward dependencies. This representation is the starting point for the decoder, which predicts one token at a time to iteratively produce the output sequence (summaries or headlines).

The attention mechanism is incorporated into the design to overcome the drawbacks of traditional Seq2Seq models. When creating each token, the attention layer allows the decoder to concentrate on the most pertinent input segments by dynamically allocating weights to various input sequence segments. By reducing the bottleneck of compressing the whole input into a single fixed-size vector, this method improves the model's interpretability and performance for lengthy input sequences.

The encoder-decoder architecture, the mathematical foundations of BiLSTM, and the attention mechanism implementation are all thoroughly examined in the dissertation. Empirical findings show how well the model captures long-range relationships in the input and generates accurate and coherent summaries. The study emphasizes how crucial it is to incorporate attention processes into Seq2Seq frameworks in order to get over conventional limitations and perform better on text summarizing tasks.

By providing insights into using BiLSTMs and attention processes for improved context representation and dynamic focus, this study advances the expanding area of deep learning-based text summarization. The results demonstrate how these methods may be used to increase the flexibility and scalability of summarization algorithms on a variety of textual datasets.